

Humanoid Robots in Indian Factories

[0:04] Good morning, everyone.

[0:06] And a happy Janmashtami.

[0:08] I think if you are on the streets of Mumbai, this is a pretty funny thing to talk about.

[0:14] What human rights?

[0:15] There are no human rights here.

[0:17] But on Monday, I came back from China, and the streets of Shenzhen, you had to see a bunch of human rights.

[0:24] So you would be hard-pressed to say no to this.

[0:31] I was in the World Robot Conference in Beijing, met around 100 plus companies building humanoids, all of different shapes, sizes, doing work from industrial to entertainment to research.

[0:46] And a stone's throw away, there was the Beijing Humanoid Olympics that were happening.

[0:55] I'm sure A lot of you have seen those funny videos of humanoid robots running and then breaking apart.

[1:05] But what was actually very interesting for me was that there were more than 2,000 humanoid robots in that stadium.

[1:15] And this race last year, it took 20 seconds to run the 100-meter dash.

[1:22] This year, it beat Usain Bolt's record by more than a second.

[1:28] That's the insane kind of development that's happening in humanoid robots.

[1:33] It broke the high jump record for humans by more than a meter.

[1:40] And that's all very good.

[1:45] But there's another side to this story.

[1:50] When you come to Real hardcore industrial floors, you know, not where all this, or your air conditioned EMS manufacturing facilities.

[2:02] I'm talking about metal forming, welding, EV automotive, wire harnesses and stuff like that.

[2:11] On such factory floors, across the whole world, there's less than 500 robots.

[2:17] Well, how do these things exist together?

[2:22] Because all the acrobatics, the backflip, the running, all of those are actually a physics problem.

[2:30] So you can solve a lot of that by using reinforcement learning in simulated environments.

[2:37] However, the factory floor is a completely different problem statement.

[2:43] And that's what we're going to talk about today.

[2:46] But before that, let me take you to a real factory So this is a general purpose bimanual humanoid robot.

[3:04] It's running an 80 ton stamping machine, producing parts for the automotive supply It wears the same gloves as what a human would be wearing on its hands.

[3:21] And it's running at 1x speed.

[3:25] This is the real speed of the robot.

[3:27] So it does the whole stamping, picks up the finished part, puts it, throws away the scrap by picking it up.

[3:40] It's also learned to use human tools.

[3:44] to actually make it work faster.

[3:59] And as time progresses, it has also learned to do multi-stage work and use its hands together with the tools.

[4:10] As you can see the dates, all of this is happening within a few weeks.

[4:19] And this was two days ago.

[4:22] This is the Manav Yumana robot.

[4:24] And for the first time, it is doing a multi-step operation.

[4:29] It is actually doing a long horizon task, where it's stamping the same plate multiple times.

[4:35] Well, metal parts are large.

[4:37] But at the other end, what's very hard to manipulate are wire harnesses.

[4:43] They're small, they're flexible, they're round.

[4:47] Here you can see the MARA robot actually crimping, putting those metal parts at the end of a wire.

[4:57] And this, again, is just a few days ago.

[5:01] This is how thin the wire is.

[5:03] I think most of you can't even see how thin this is.

[5:06] This is a 0.5 square millimeter wire.

[5:08] It's learned to manipulate those thin wires as well.

[5:12] And this is what makes it so exciting, that within just three months from us not being able to pick up a plate, we are now able to manipulate such fine objects.

[5:28] Out of those 500 odd robots on real factory floors, 10 are ours here in India.

[5:35] So, yeah, I think humanoids are available a little slow here in India, but let's get into that.

[5:43] And before that, this is just quickly our robots are powered in a wheel base and in a fixed base.

[5:50] We don't do fancy bipedal robots.

[5:54] We make robots which are highly reliable.

[5:58] and then survive the 40 degree summers here on Indian factory floors.

[6:05] So a few months ago when Venhu and I started chatting about robotics, he posed this question to me that what is going to be different about the physical AI race as compared to the LLM race?

[6:20] We kind of lost that, huh?

[6:24] And so what will it take to participate in this race?

[6:28] Is it national scale compute?

[6:33] Or is it building very robust hardware supply chains?

[6:39] Or is it something else?

[6:41] If you ask me, it is something else.

[6:43] But let's just get the compute and the hardware out of the way.

[6:50] So, In LLMs, you are now training at the frontier of this trillion parameter plus models.

[6:58] You need millions of GPUs.

[7:00] You need national scale capital to make that happen.

[7:05] However, the action flow is very different.

[7:08] When a bot is slipping out of your finger, you cannot have a data center running a trillion parameter model telling you what to do.

[7:16] You have to react within a few milliseconds.

[7:20] Which means you have to run these models at the edge, which, for the foreseeable future, makes these models in the sizes of a few 10s of billion parameters.

[7:33] So to train such models, you need thousands to tens of thousands of GPUs, expensive, but definitely doable in India.

[7:45] Next, the hardware supply chain.

[7:48] But to be honest, every company on earth is subscale.

[7:53] So all of our supply chain is transferred to China.

[7:56] Let's be honest about it, ours does as well.

[8:00] If you really break down a robot, it's a bunch of actuators, connected by mechanical linkages, controlled by AI compute, and fed by some power parts.

[8:18] So the mechanical and the power electronics parts are kind of sorted because of the last 10 years of work in the supply chain.

[8:30] Compute, well, that's a problem.

[8:34] Robotics is not unique in that problem statement.

[8:36] There's a lot of work happening there.

[8:39] So I'm not going to that.

[8:41] What is unique about robotics are these actuators.

[8:44] Those are all the joints that actually make the robots work.

[8:49] And if you break the actuator down, it's two critical parts.

[8:55] It's very high precision manufacturing for the gears and rare earth magnets inside your motors.

[9:06] Just like you know the EV supply chain with scale and time has been built in India.

[9:12] I believe that with scale you can build this supply chain as well.

[9:16] It will take time but it's definitely again doable.

[9:22] So now that like The US, with its trillions of dollars in compute, has trained models, as Venkat said in his presentation, which can cook, which can clean houses, but not something that can work on this very hard and dirty, dangerous floors of, let's say, a metal forming factory.

[9:55] China, with its huge manufacturing scale of robots are still deploying most of these humanoid robots in data collection centers or in terms of doing material movement, warehouses and stuff.

[10:11] So why is that number 500?

[10:15] Because the fundamental constraint for physical AI is actually physical data.

[10:22] And it's very different from the LLM base, because this is not just videos on the internet that you can scrape and you can build into a data set.

[10:38] Ultimately, the work that happens in a factory floor happens not just with vision, but also with touch.

[10:45] Because when a cable connector locks, or when a part slips out of your hand, you don't see it, you feel it.

[10:54] Ultimately, this muscle memory of workers is not digitized.

[11:00] There is no internet equivalent.

[11:03] And that is the part where India can really shine.

[11:08] So I'll just quickly touch upon the pyramid with which your data is collected in robotics.

[11:17] So the first is internet video, and it is possibly the worst kind of data that exists.

[11:27] There's a lot of research, but it's very hard to take internet video data of people doing things and translating it into robot actions.

[11:38] What has got a lot of attention now is egocentric video capture.

[11:43] And unfortunately, here in India, we have found it very easy to strap these GoPros on people's heads and record all of this and sell it to the US at \$7 per hour.

[12:01] So, thank you for selling gold at the price of peanuts to the world.

[12:07] However, this kind of video data, I think this is actually extremely an innovative space and there's a lot of research happening.

[12:19] This kind of video data has been used in all of the largest models that have come out at the frontier.

[12:29] So these three companies, all unicorns in the US, have all released models in August, doing a bunch of different things for whom work.

[12:40] What is very interesting is that a very large part of their data has come from India in order to make this happen.

[12:49] But the video learning works at the home where you don't need millimeter level precision.

[12:55] If you want to work on a factory floor, you need that.

[12:59] And that's where China realized a few years ago that you need to collect data with real robots.

[13:09] I'm sure many of you have seen these videos of these massive data factories.

[13:17] tens of thousands of square foot, where you have hundreds of robots just collecting data day in and day out.

[13:27] And all of this paid for by the Chinese government because they're creating a national asset.

[13:34] And not an hour of this is allowed to leave China.

[13:39] So what is at the top of this pyramid?

[13:44] Well, you don't deploy robots in those data patches.

[13:48] So for your post-training, the most valuable kind of data that you have are basically data from real robots doing real work on real factory lines.

[14:00] And that is where we are concentrating.

[14:03] And that's where I feel India has a definite part to play.

[14:11] Why?

[14:12] Two main reasons.

[14:14] First, diversity.

[14:17] India has amazing diversity when it comes to manufacturing across different industries at different levels of automation.

[14:28] And that allows us to deploy robots in these varied conditions, some of the most challenging factory floors on Earth.

[14:38] And that is exactly the kind of data that trains really good models.

[14:45] The second is the youth people, and it's a very interesting fact that we have learned the hard way.

[14:54] So, in the beginning, robots need robot pilots to operate them.

[15:01] Now, you can't just have anyone become a robot pilot, because we realize that the data that comes from some humans is inexplicably much better than others.

[15:14] So we train reward models that can see a human performing the job of a robot pilot, operating, teleoperating these robots, and the reward model can tell us that you know this person is actually a good robot pilot versus the other and we don't know why like we can't explain it so this entire AI hiring harness for us just in the month of August has processed more than 15,

[15:45] 000 people through visual motor tests, spatial reasoning tests, simulated robot runs, real robot runs to

[15:53] create this scheme of robot pilots for us.

[15:57] Robot pilots, you know, a job that did not exist like a few months ago here in India.

[16:03] But I believe it's going to become one of the most lucrative new opportunities for the Indian youth.

[16:11] And putting these two together, I believe we in India can become the missing internet for the whole physical AI race.

[16:26] You know, as a company, we'll obviously keep pushing the frontier, but the ecosystem also needs to come together around this.

[16:36] Three key things.

[16:37] First, just make it easier to build robots in India.

[16:43] Have PLI-like schemes to make it much easier for us to do this.

[16:50] Second, make it lucrative for the industrialists to deploy India-made robots.

[16:57] At Ola Electric, we saw how helpful fame was for us.

[17:00] Like, there should be a fame equivalent for India-made robots as well.

[17:06] But most importantly, India, the champion for building visible public infrastructure, should figure out how to keep its data as a national asset.

[17:21] I'm not saying that we like China and ban selling of this data, but at least keep a copy of it here in India so that any company that's training physical intelligence has access to it.

[17:38] So I'll go back to the factory floor and you know, Something that we saw in the beginning, this is what makes me so naively optimistic that in three months, if our 10 robots can learn more than 30, 35 tasks, going up the level of difficulty.

[18:04] Like even last night, after I reached here, I was on a call with my co-founder that one of the factories where we had deployed, They produce parts for Toyota's four-wheelers.

[18:18] So we have never operated a stamping machine which is more than 100 tons.

[18:23] But this factory promoter saw the robots and he's like, you know, you guys are doing so well on this.

[18:30] There's this 250 ton stamping machine.

[18:33] Why don't you go and try your hand at that?

[18:35] I'm sure your robot can do it.

[18:37] So this kind of optimism from the industrialists The appetite to try on this new technology makes me naively optimistic.

[18:55] You know, this technology is not just about, say, whatever, this 10 robots today, 100 robots at the end of the year.

[19:04] This is about the next unit of labor.

[19:09] And the world recognizes that.

[19:10] Now you have all heard the US banned Chinese humanoids.

[19:15] So we also need to wake up to this and move fast.

[19:20] Over the last three years, we kind of lost the LLM race because we couldn't build the compute.

[19:27] But we can build the compute.

[19:31] We could not harness the internet's data in that race.

[19:35] But in this race, The data will be produced by hands that's part of this country.

[19:42] So we should own it.

[19:44] And the third and most important thing is that it's easy to put your LLMs out there in the world, right?

[19:54] I mean, weights can travel on the internet.

[19:56] But it's not so easy to deploy robots.

[19:59] First, they need to clear customs.

[20:01] So...

[20:03] I will take one of Vinu's statements and I'll change it a little bit.

[20:07] I think India needs its unity more than it needs its deepseek.

[20:15] Okay, that's it.

[20:16] Thank you.

[20:27] Thanks so much.

[20:29] So I forgot to mention that there is a QR code in your...

[20:32] where you can actually put in questions.

[20:36] And of course, I wrote a couple of questions, unless any of you want to ask questions.

[20:44] I'll just start, we still have 10 minutes for Q&A.

[20:47] So Sudha, thanks a lot, very interesting presentation.

[20:50] You know, broadly I want to understand your landscape as two broad questions.

[20:56] The only company humanoid equals industrial robots that I have encountered is Hume.

[21:03] Is it working?

[21:08] Yeah, see, I think a lot of what we do in terms of training frontier robot models.

[21:29] is in the realm of science, it's not yet in the realm of technology.

[21:33] And as you've seen in the past, India is great at taking technology and packaging that into products.

[21:41] So I think a little bit of that overhang is there in the robotics industry as well.

[21:46] There are a bunch of interesting companies which have been doing robotics in India for a while.

[21:52] Some of them have been deploying for warehouse automation.

[21:57] Uh, a little bit more in terms of like industrial uses, but in terms of training this kind of frontier intelligence models, it's pretty new.

[22:08] There are a few companies that you can find around Bangalore.

[22:11] But also what's very interesting is to see a lot of very young companies take on this challenge and build very unique products.

[22:21] I'll actually name one of them.

[22:23] The founder was actually part of my physical AI team at Ola.

[22:28] So they built this company called iFind Robotics.

[22:32] It's almost like taking these Disney characters and making them real.

[22:37] So I think there's a lot of very diverse approaches towards robotics, some to making these robots feel very approachable in the home that you can play with, and others at the extreme end like ours, where you're building robots, which are very hardcore working on factory lines.

[22:58] But I'm seeing more and more of these companies come out, colleges like IIT Bombay, a bunch of these places which are doing something out there.

[23:13] I've got a bunch of interesting questions, so it looks like a lot of curiosity around what you guys are doing.

[23:19] So I'm going to pick up this and probably just explain.

[23:23] One of the things that actually came to mind is that when you look at India, and especially the industrial practice, the perception usually is that the cost of labor in India is extremely low.

[23:36] If you actually do assembly in India, depending on which part of India, it varies from 8,000 to 12,000 rupees per month.

[23:44] That's the kind of salary they get.

[23:46] If you look at some of these units you're operating, they seem to be just more technical handling work than anything else.

[23:52] So the question that is also coming in is more centered around this, what is the economics of this business, why would an industrial house want to have a lot of such robots in their facility?

[24:07] So I think first of all, when you talk to a promoter owning, say, a metal forming factory, if you ask them, what is your biggest challenge, they're going to say, labor to

work in our factories.

[24:23] Understand that one of the use cases that we are deploying in has some of the highest attrition.

[24:30] Working on a metal forming floor is extremely disturbing for humans.

[24:40] It's too loud, there's a lot of smoke, your body gets challenged on a daily basis, meaning that you find people churning within three, four months of starting that work.

[24:51] Now, your economics of employing these people, the salary is actually a little bit higher.

[24:56] It's in the 20 to 30,000 range.

[24:59] And the fully loaded cost is actually much higher because you need to keep training them, you need to backfill them.

[25:06] But ultimately for these factory promoters, what's most important is the reliability of that labor.

[25:14] So when you add up the cost of human-centered mistakes, doing such gull work.

[25:23] Our factory owners are seeing break even just on wages plus just lost materials within 18 to 24 months.

[25:33] And that's the reason why we are seeing so much pull from the industry.

[25:39] That being said, it's very early on for us and with scale, you know, there's this that with every change of magnitude, the cost halves.

[25:51] And that's exactly what we're looking at.

[25:53] So when we're in the like 10,000, 5 to 10,000 scale, that's where it makes a lot of sense for.

[26:06] A lot of on the cost side of things.

[26:09] So I'll just mix those questions in one sentence.

[26:12] So finally, So, the way I kind.

[26:32] Of broke down the robot, the actuators are basically the largest part of that cost.

[26:42] You can say roughly robot would cost around say \$20,000 to make.

[26:48] Out of that, 50% would basically be the cost of actual testing.

[26:56] And the rest of the three components, the mechanical components, the energy components, and compute, say it is like equally divided.

[27:05] It's not that I'm just simplifying it.

[27:08] So what we foresee is that With scale, the cost of actuators is going to come down significantly.

[27:17] Because the other components themselves are more or less like, you know, again, because of the EV supply chain and stuff like the cost is actually pretty low.

[27:30] Those will also come down because you're making things specifically for robots.

[27:34] But we feel that over the next few years, The cost of such robots, making such robots would come down to below \$10,000, where a large part of it will actually become compute.

[27:51] The percentage of the total cost of compute is actually going to keep increasing because the model sizes are going to become larger and larger.

[28:01] So you will have to keep putting bigger compute boxes that NVIDIA will send to you as part of these robots.

[28:10] But yeah, that's high level feeling.

[28:21] Yeah.

[28:22] So again, this is something which is very interesting about India.

[28:27] You would always assume that technology like this, people would want to take it as a service, like robot as a service.

[28:35] Now when we talk to these promoters, they're like, you know what, when you're doing your POC, you please do it free of cost.

[28:42] Because there's something entirely new.

[28:44] You're going to put up my machines, you're going to waste a lot of my raw materials.

[28:48] That is how I will help you, but you will do the proof of concept at your cost.

[28:53] Like, okay, fine, we'll do that.

[28:55] But the moment it starts showing value, At that point, they don't want to take it as a service.

[29:02] They want to buy it because they are like, you know, you're not going to have one margin if you give it to me as a service.

[29:07] So outright sales is what we are seeing as a model for deployment.

[29:12] And like I said, depending on the use case, the ROI is somewhere between 18 to 24 months for these deployments.

[29:25] I'm saying quite a few questions.

[29:29] So, in terms of your application, what is it about?

[29:50] So, today we design these robots in India, all of our components are sourced from China.

[30:00] 951, and maybe a single them here in India.

[30:03] And, uh, for the foreseeable future-say, for the next year or so-I don't see that

changing much.

[30:11] There's obviously a bunch of players outside of China who become pretty relevant, so you would think of them as tier-1 auto suppliers from Korea, from Germany, who would have similar technologies built for EVs, which you can miniaturize and bring, uh, to robotics.

[30:33] So right now they're not very cost-effective, but given the world's push to find supply chains outside of China, they are becoming more and more relevant.

[30:47] You know, one of the curiosities I have is: when I look at your own model and the shipwright, when you get started in business, what do you want to be?

[30:56] Do you want to be a humanoid robot equaling manufacturer, who will supply you for any industrial and non-industrial needs?

[31:04] Or was it that, you know, it doesn't matter; physical devices come from anywhere, I'll just train it?

[31:09] So what is the broader ambition, and more importantly, if you could just share just yourself an idea of scale, how much money they will take, you know, for a greater business like this, and of course there's an upfront investment and overtime you'll be spending, so let's say 84 billion dollars in revenue.

[31:30] So, uh, see, when we started the company, our whole-and-sole mission was to make this kind of dull, dirty, and dangerous work in industrial factory floors optional for Indians.

[31:44] Like, that was the mission with which we started.

[31:47] And that has now become a number for us, that, hey, in the next 5 to 6 years, we're going to deploy 100,000 such robots where we don't have people doing such backbreaking work.

[32:01] Now, uh, that means you will have to build the robot bodies, that means you'll have to train the models, that means you'll have to collect the data in order to train those models.

[32:14] So whatever is needed to make that mission successful, we will do.

[32:19] Like, you know, that's how we have approached the problem statement.

[32:24] We've realized that you can't just take a Chinese robot and take an American brain and put it together on a factory floor in India.

[32:34] Those are the learnings of the journey.

[32:37] So, uh, each part of the problem that needs solving would have to be solved.

[32:43] And I'm sure with time India would have a much bigger, flourishing humanoid sector with companies solving different parts of that challenge.

[32:54] For us, as one of the first movers in this, we've had to solve a lot of this.

[33:00] Now, in terms of a billion-dollar revenue-well, uh, if I have to kind of put the numbers together, so maybe I'll do 100 million revenue first, that is easier.

[33:13] That's the part that we are chatting about.

[33:17] So, you know, we're not yet chatting about the billion-dollar part, maybe in some Excel sheet for our investors in there.

[33:27] But, uh, 100 million dollars in revenue is basically 5,000 robots on the line.

[33:33] So it will take you, uh, in the magnitude of a few hundred million dollars to get a business like this off the ground.

[33:43] But obviously your operating leverage improves as, you know, you can set up much bigger manufacturing setups, better supply chains beyond that.

[33:55] But it is not, uh, incomprehensible to build such a business in India.

[34:00] It is not the kind of hundreds of billions of dollars that, say, the LLM companies have raised in the US in order to build such a business.

[34:14] Right.

[34:14] Do you have a couple of questions around, broadly, you know, data and training

as well?

[34:20] So things like, you know, these aspects of, you know, training.

[34:25] Most importantly, it seems to me that, you know, training a particular robot for a particular application-so how general-purpose can it be?

[34:36] So the data that we collect is used in different parts of training.

[34:41] Ultimately, the vision-contact-action model that runs on our robots, it's understanding the primitives of manipulation.

[34:51] And, uh, that's the reason why, from its learnings in the metal-forming industry, suddenly we see a lot of transference into wiring harnesses, even though, you know, there's nothing in common between them.

[35:08] Apart from the fact that it needs to understand the basic physical causality of slipping, jamming, and things like that.

[35:18] So the data that we collect is annotated not just from a vision perspective but also from a tactile data perspective, and those teach our models the basics of manipulation.

[35:33] So these models are not, like, you know, trained for just, say, one task that we're doing for one factory floor.

[35:42] You have a larger model, and then you fine-tune it for deployment on each of these tasks.

[35:49] So the more tasks that we do, the higher diversity of applications that we have actually makes us stronger and better.

[35:59] What was the other question?

[36:01] In terms of safety.

[36:03] Okay.

[36:03] So I'm assuming that this is safety, not from, you know, the robot going broke.

[36:10] That's an alignment problem.

[36:12] We're not there yet, so there's not much work that we're doing on that front.

[36:19] But in terms of actually the robot operating in a safe space, we actually mandate that these workstations that we have, say, these tapping machines, or the setup of the wiring harnesses, is done in a separate space on the factory floor.

[36:38] So we don't coexist where humans are doing their work right now.

[36:43] Thankfully, because our robots are on fixed bases or wheel bases, the chances of them falling down are much lower.

[36:52] And a lot of safety is actually very well explored in the realm of robotics for, like, you know, the past few decades.

[37:02] So the AI model itself runs inside what you can think of as a box of heuristics, which prevents certain actions, certain speeds, certain motions from happening.

[37:15] So while the model might say, "Do this," those safety parameters actually limit the robot from doing such actions.

[37:26] You know, I kind of let it come down to, uh, can India benefit from the one of the things that you also asked?

[37:33] What does it change to build that kind of a model?

[37:36] Because it actually also means it might-the data, of course, is going to really need something that looks like a human, a walk like a human.

[37:45] But that debate aside, the point is, if you want to recreate something that you can create, the only way India does that is by creating.

[37:53] Is it possible?

[37:54] Yes.

[37:55] Given where the world is today, I think it's very much possible.

[38:00] And, you know, that's the reason why we're so optimistic about building this venture out of India.

[38:07] I think what needs to happen is a lot more work from the ecosystem together.

[38:12] You know, everyone's hopeful of seeing some robotics action in the next budget.

[38:18] So once we see all of that coming together, I think it's very much possible.

[38:24] And unlike a lot of feeling in the room that, you know, what will industrialists do, I understand that each one of them have gone to China, they have seen these robots, but they don't want to bring a Chinese humanoid robot and implement them on their factory lines.

[38:44] So they are extremely bullish about taking these general-purpose robots, taking these foundational AI models, and implementing it here.

[38:54] Because ultimately it's not just about the factories that get built.

[38:59] It's about, you know, so many of these new factories that are coming up.

[39:04] Like, for a lot of promoters working with us, they're like, you know, you do your POC here, but you know what, I'm building this next factory for this big use case.

[39:16] Let's make that lights out, let's make that a full-dark factory, so that it hits that kind of cost and quality benchmark that took all of this manufacturing away from India to China in the first place.

[39:32] So that's what makes this so exciting, and that's why I think it is very much doable in India.

[39:38] So I have two last questions.

[39:40] So one is, getting a bit deeper into your own revenue model.

[39:43] You mentioned 5,000 robots on the floor, you can make 100,000 per year.

[39:47] So how does the revenue model work, do you think?

[39:49] So because you're actually investing in building the physical robots for training in Fortinet, so you actually sell one dedicated device to industry, or do you have recurring service revenue streams coming from there?

[40:01] So, uh, it's a mix of both.

[40:03] Initially it's the hardware sale.

[40:06] Obviously the margin on that is not very high.

[40:09] But you will operate these robots for a minimum of 5, 6 years.

[40:14] Like, you know, the robotic arms today, the ABB KUKAs of the world, people use them for, like, 15, 20 years.

[40:22] But we are now, you know, building models, estimating around 5 years of such usage for these robots.

[40:30] So there is recurring revenue that comes both from servicing of these robots as well as from AI updates.

[40:38] Because each update actually makes these robots more productive, whether from that single task that they're doing, or as a lot of use cases we're seeing, where you use the same robot, say, during the day for one use case, during the night for another use case.

[40:58] Like, we actually have one customer who uses the robots for stamping during the day, and at night it actually does this box printing, basically, for their logistics and stuff.

[41:11] The same robot can do both.

[41:13] So as we add new skills, we make it more productive.

[41:17] There's recurring revenue from AI as well.

[41:21] Great.

[41:21] The last question for me is more a question centered around your own journey of starting and scaling up the company.

[41:27] My understanding is, from what I have discussed with you in the past, that your bootstrap is, you know, VCP funding has been taken.

[41:34] So it's very, very interesting.

[41:35] You've reached this stage without any funding.

[41:38] So what I want to know is, what are the next steps for you?

[41:41] What is your bottleneck that you see when you scale up your business, and more importantly, what is your own funding journey from here?

[41:48] Yeah, I think building Honda Electric obviously has taught us to be very capital-efficient.

[41:56] So me and my co-founder, like, we've been there for almost a decade in the company.

[42:03] So it has given us the freedom to actually explore what general-purpose robotics could mean in India.

[42:11] So we kept it bootstrapped.

[42:13] Some of the top entrepreneurs in India have been a part of that journey with us.

[42:19] In terms of capital, I think as we are now kind of going from the 10 robots to 100 robots by the end of this year, and to 1,000 next year, we will be picking up institutional capital in order to do that.

[42:36] But it will be on the back of decently large order books and revenue streams.

[42:42] So we don't want to build a business where, you know, we raise, like, a huge amount of capital and revenues, like, very far away.

[42:52] But it's pretty interesting in the world of robotics that you don't need to train very, very large models to make these robots economically useful for people.

[43:05] And that's been our focus to grow sustainably, but also scale up really fast.

[43:11] Because only with scale do you see the data flywheel coming in.

[43:17] And ultimately that is what is going to determine whether people use Indian physical AI brains, or, you know, they have to use the ones from the US.

[43:28] Thanks.

[43:29] Thanks for the interview.

[43:32] I'm going to do the intro of the thing we're going to do.

[43:38] So whenever you're ready.

[44:07] Just to check, is the mic is on?

[44:09] Wonderful.

[44:10] I think.

[44:11] Fantastic session on robotics in the factory.

[44:14] We will now switch gears a little bit and jump to a very interesting, complex second topic: healthcare.

[44:22] And in healthcare we have speaker today with us, Shanku Waria.

[44:27] He's the founder and CEO of Cure.ai, a global AI company deployed in more than 105 countries and innovative solutions for healthcare.

[44:37] A technologist and entrepreneur, he has built Cure.ai into world's one of the most recognized AI innovators, with solutions that cover early detection and management of lung cancer, neurological findings, and specific diseases, to support clinicians and proper development.